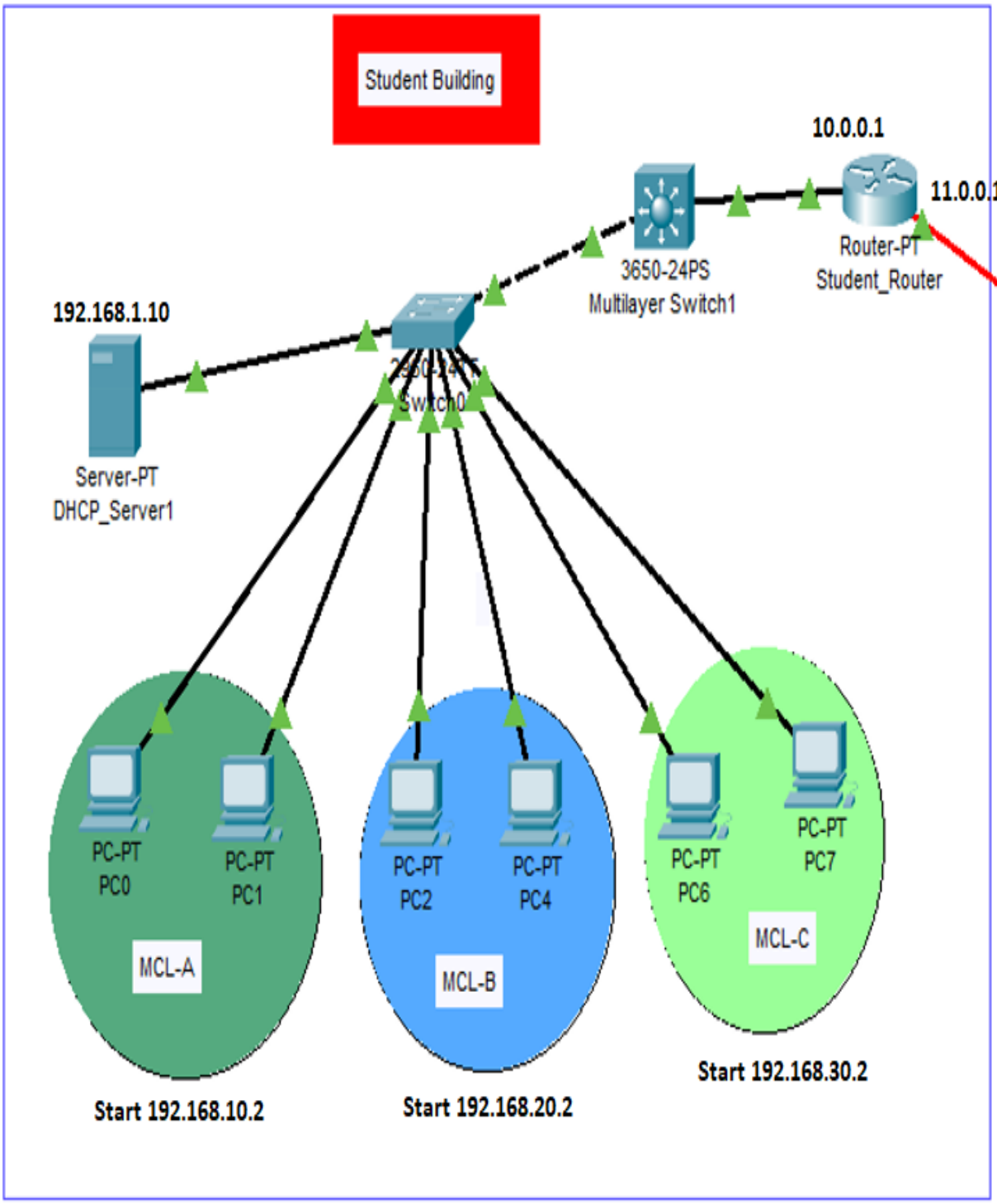
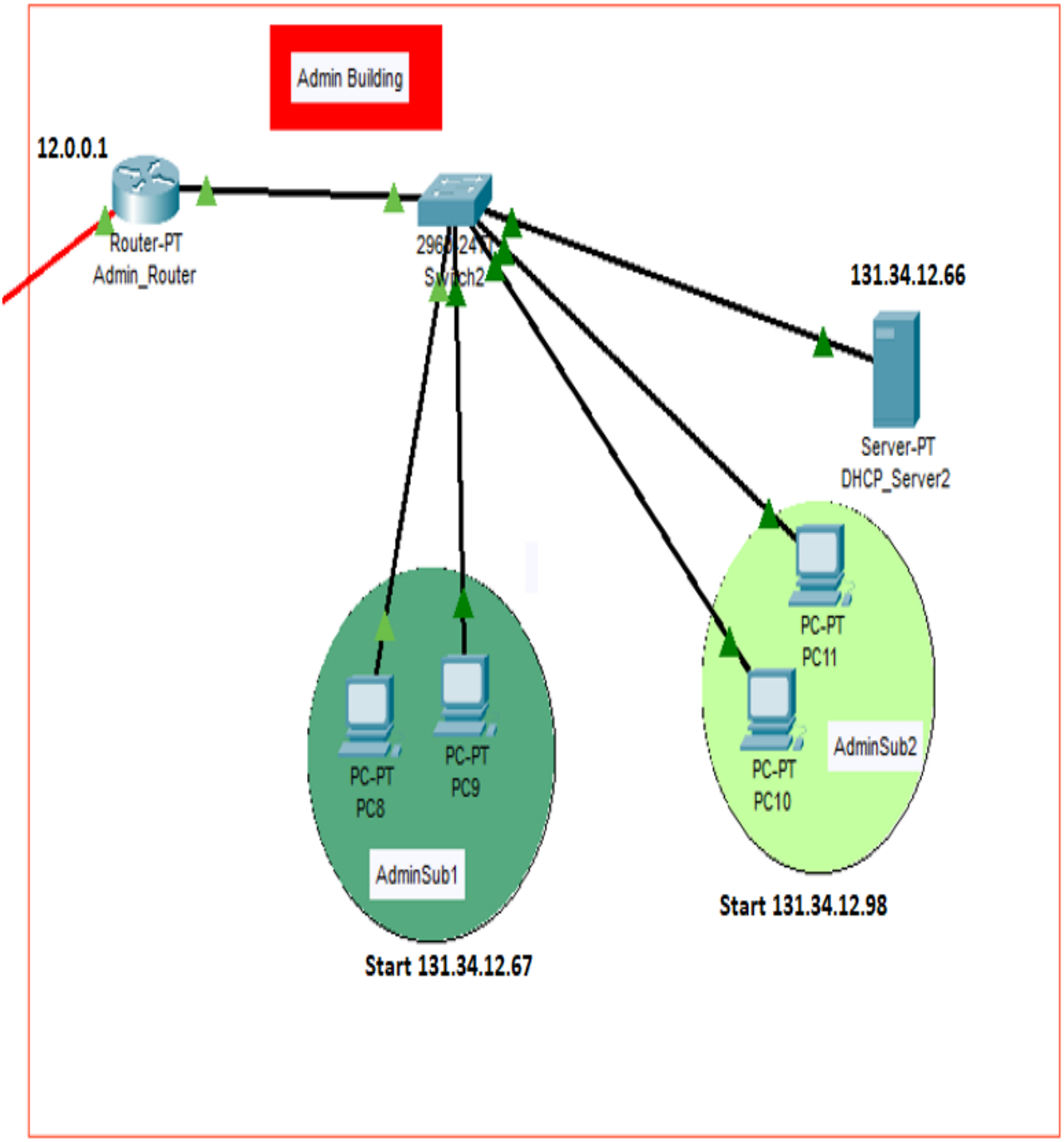
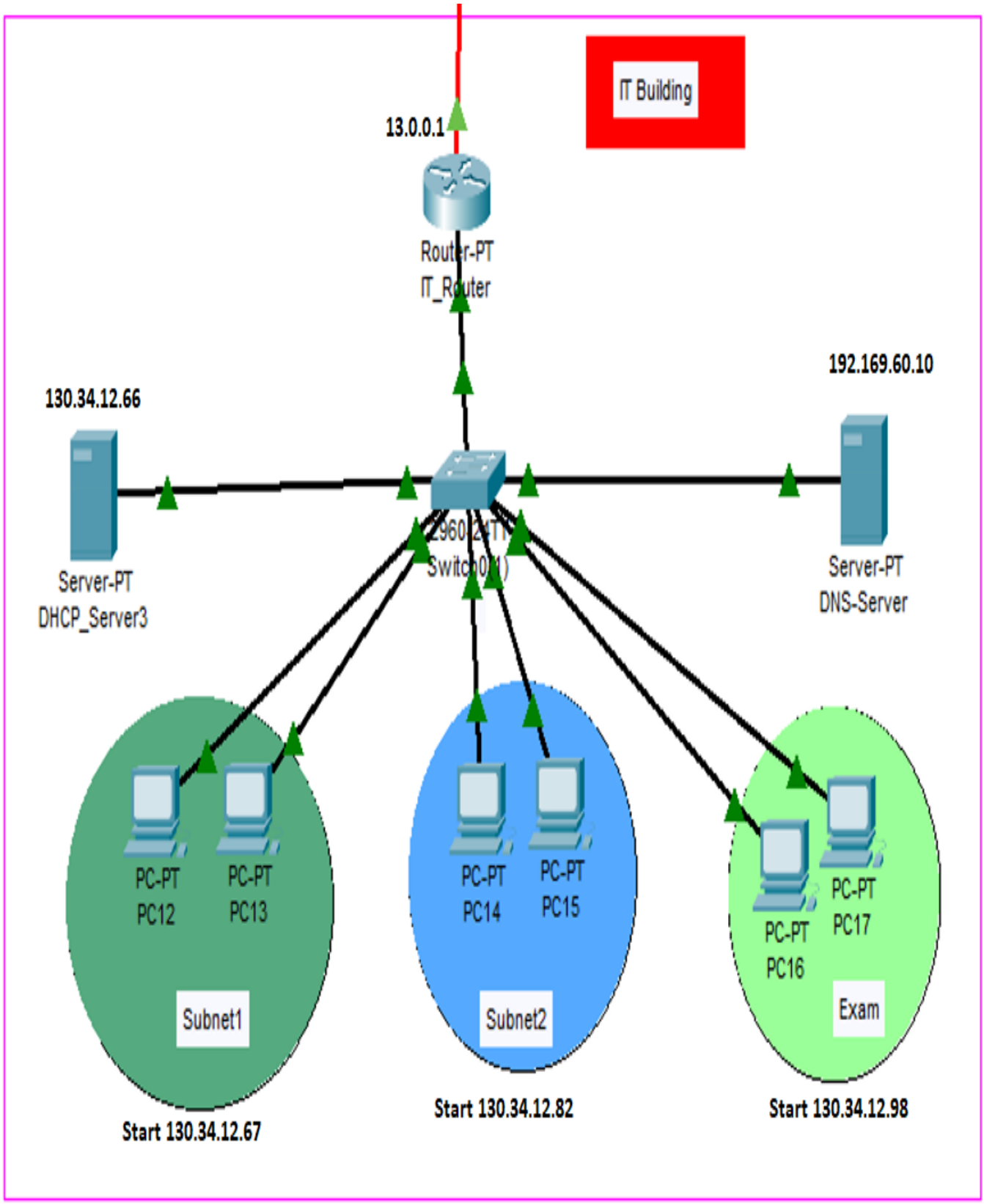
Inter building Routed Network Design




Admin Building
12.0.0.1
Router-PT
Admin_Router
Switch2
131.34.12.66
Server-PT
DHCP_Server2
PC-PT
PC11
PC-PT
PC10
AdminSub2
Start 131.34.12.98
PC-PT
PC8
PC-PT
PC9
AdminSub1
Start 131.34.12.67


IT Building
13.0.0.1
Router-PT
IT_Router
192.169.60.10
130.34.12.66
Server-PT
DHCP_Server3
Server-PT
DNS-Server
PC-PT
PC12
PC-PT
PC13
PC-PT
PC14
PC-PT
PC15
PC-PT
PC16
PC-PT
PC17
Subnet1
Subnet2
Exam
Start 130.34.12.67
Start 130.34.12.82
Start 130.34.12.98

IP Address Table

Device	Interface	IP address	Subnet Mask
Student Router	Fa0/0	10.0.0.1	255.255.255.0
	Se2/0	11.0.0.1	255.255.255.0
Admin Router	Fa0/0.1 (AdminSub1)	131.34.12.65	255.255.255.224
	Fa0/0.2 (AdminSub2)	131.34.12.97	255.255.255.224
	Se2/0	12.0.0.1	255.0.0.0
IT Router	Fa0/0.1(Subnet1)	130.34.12.65	255.255.255.240
	Fa0/0.2(Subnet2)	130.34.12.81	255.255.255.240
	Fa0/0.3(Exam)	130.34.12.97	255.255.255.240
	Fa0/0.4(DNS Server Network)	192.169.60.1	255.255.255.0
	Se2/0	13.0.0.1	255.0.0.0
Route_Router	Se1/0	11.0.0.2	255.255.255.0
	Se1/1	12.0.0.10	255.0.0.0
	Se1/2	13.0.0.2	255.0.0.0

Student Building VLAN Assignment

VLAN	DHCP Start IP
MCL A	192.168.10.2
MCL B	192.168.20.2
MCL C	192.168.30.2

Admin Building Subnet Assignment

Subnet	DHCP Start IP
AdminSub1	131.34.12.67
AdminSub2	131.34.12.98

IT Building Subnet Assignment

Subnet	DHCP Start IP
Subnet1	130.34.12.67
Subnet2	130.34.12.82
Exam	130.34.12.98

Student Building RIP Routing Table

Device	Neighbor Network Address
Student_Router	10.0.0.0
	11.0.0.0
Multilayer Switch1	10.0.0.0
	192.168.10.0
	192.168.20.0
	192.168.30.0

Admin Building OSPF Routing Table

Router	Neighbor Network Address	Wildcard Mask
Admin_Router	12.0.0.0	0.255.255.255
	131.34.12.64	0.0.0.31
	131.34.12.96	0.0.0.31

IT Building EIGRP Routing Table

Router	Neighbor Network Address	Wildcard Mask
IT_Router	**130.34.12.64 (Subnet1)**	**0.0.0.15**
	130.34.12.80(Subnet2)	**0.0.0.31**
	130.34.12.96(Exam)	**0.0.0.31**
	13.0.0.0	**0.255.255.255**
	192.169.60.0	**0.0.0.255**

Redistribution Point Route_Router's Table:

Protocol	Redistribution Protocol
EIGRP	redistribute rip metric 10000 0 255 1 1500
	redistribute ospf 1 metric 10000 100 255 1 1500
	network 13.0.0.0
OSPF	redistribute rip subnets
	redistribute eigrp 100 subnets
	network 12.0.0.0 0.255.255.255 area 0
RIP	redistribute eigrp 100 metric
	redistribute ospf 1 metric 1
	network 11.0.0.0

NAT configuration summary:

At **Admin Building Dynamic NAT** is configured on **Adminsub1** to access external resources and internal private IPs of **Adminsub1** are translated to a **pool of public IPs** (**12.0.0.1 -- 12.0.0.9**) .

```
Admin#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 12.0.0.1:6        131.34.12.68:6     130.34.12.83:6     130.34.12.83:6
icmp 12.0.0.1:7        131.34.12.68:7     130.34.12.82:7     130.34.12.82:7
icmp 12.0.0.1:8        131.34.12.68:8     130.34.12.83:8     130.34.12.83:8
icmp 12.0.0.1:9        131.34.12.68:9     130.34.12.99:9     130.34.12.99:9
icmp 12.0.0.2:1        131.34.12.67:1     130.34.12.68:1     130.34.12.68:1
icmp 12.0.0.2:2        131.34.12.67:2     130.34.12.68:2     130.34.12.68:2
icmp 12.0.0.2:3        131.34.12.67:3     130.34.12.67:3     130.34.12.67:3
icmp 12.0.0.2:4        131.34.12.67:4     130.34.12.67:4     130.34.12.67:4
icmp 12.0.0.2:5        131.34.12.67:5     130.34.12.82:5     130.34.12.82:5
icmp 12.0.0.2:6        131.34.12.67:6     130.34.12.68:6     130.34.12.68:6
```

ACL applied (IT_Router):

access-list 101 deny ip **192.168.0.0** 0.0.255.255 130.34.12.96 0.0.0.15

access-list 101 permit ip any any

Here, **192.168.0.0** is the network address of Student Building's 3 vlans :

MCL A 192.168.10.2

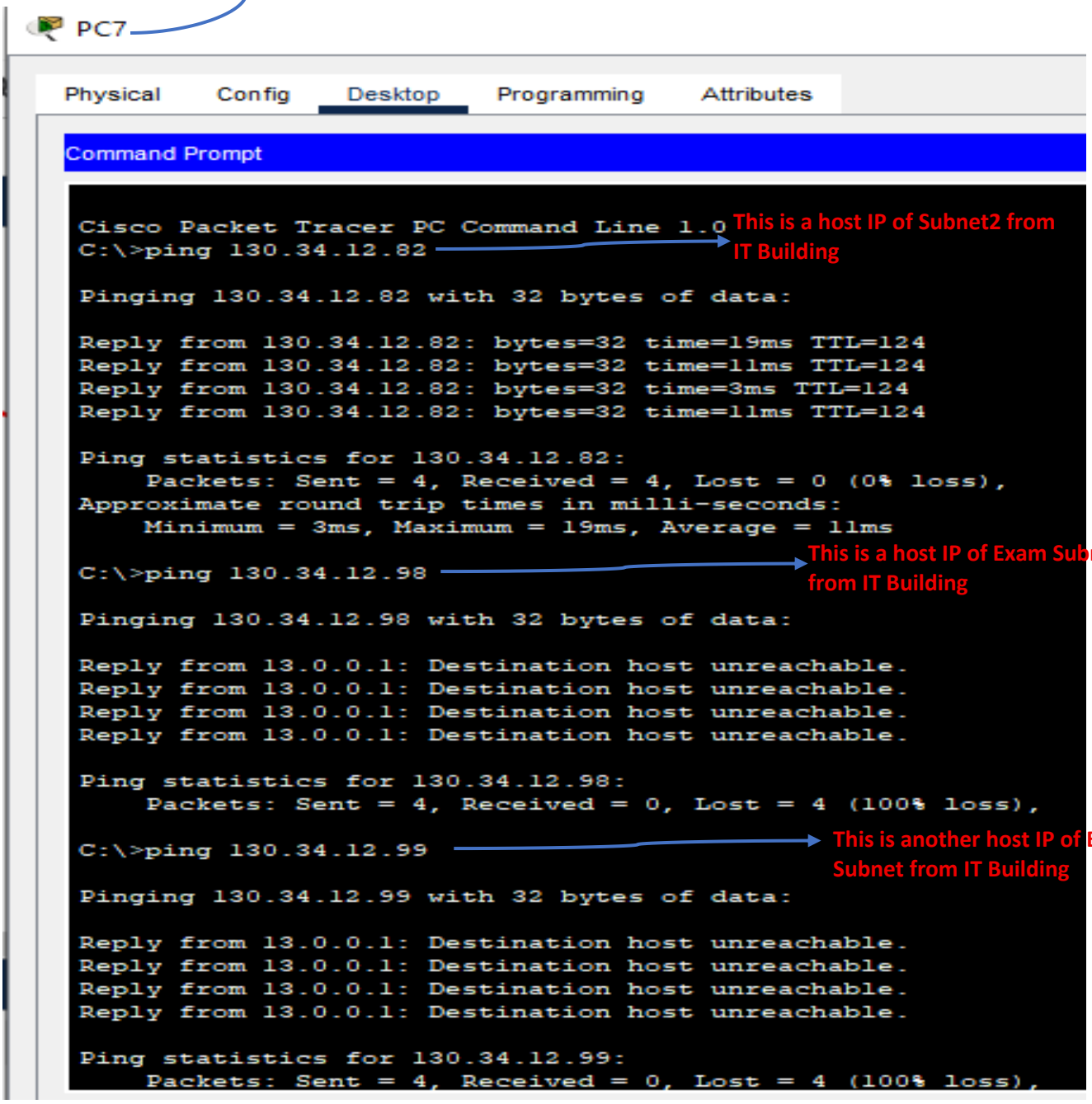
MCL B 192.168.20.2

MCL C 192.168.30.2

130.34.12.96 is the network address of **EXAM** subnet .

Justification:

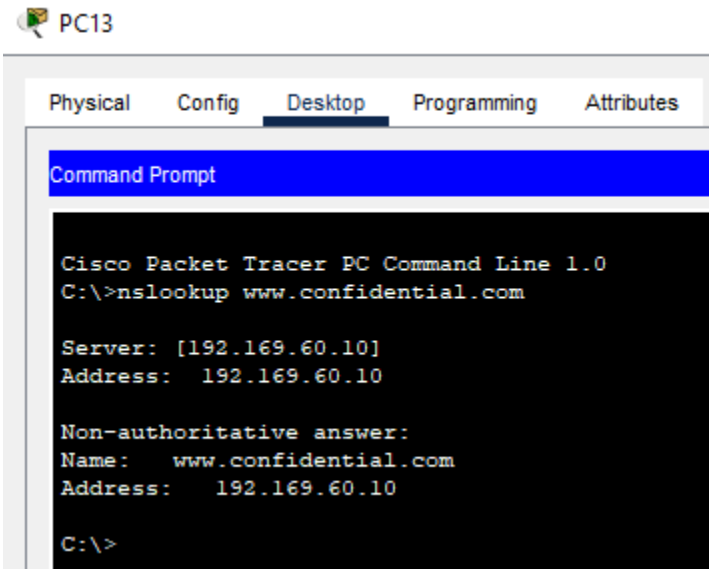
PC7 is a host of Student Building



DNS Configuration (IT Building):

DNS Server IP Configuration :

IPv4 Address	192.169.60.10
Subnet Mask	255.255.255.0
Default Gateway	192.169.60.1
DNS Server	192.169.60.10



SSH Password: admin123

